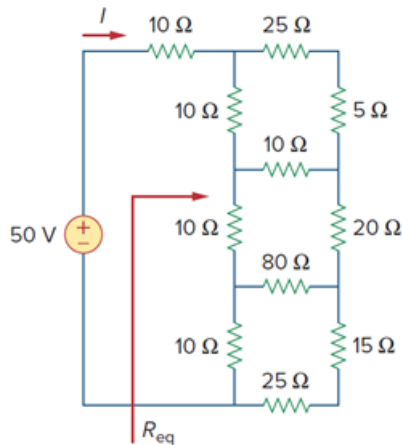


Problem

Find R_{eq} and I in the circuit of Fig. 2.121.

Figure 2.121:



Step-by-step solution

Step 1 of 7

Refer to Figure 2.121 in the textbook.

Redraw the circuit diagram by replacing the series resistances $25\ \Omega$ and $5\ \Omega$ at the top with $30\ \Omega$ resistance, $15\ \Omega$ and $25\ \Omega$ resistances at the bottom with $40\ \Omega$ resistance. Mark the upper delta as abc .

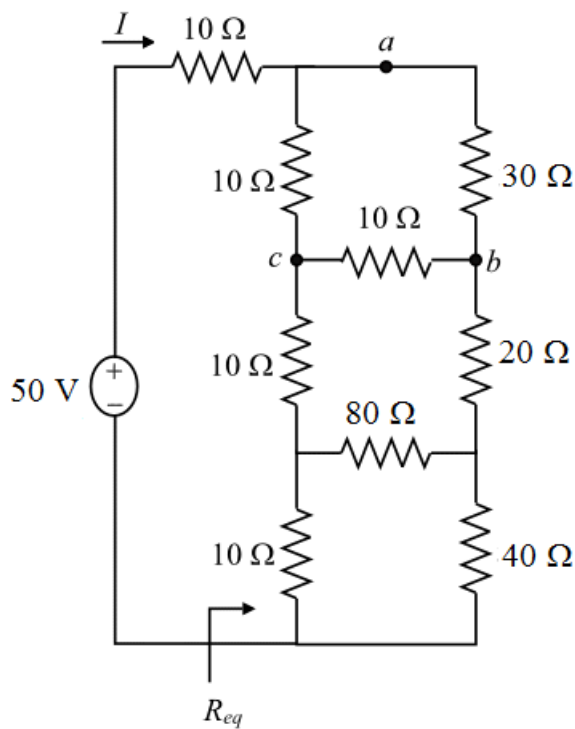


Figure 1

Step 2 of 7

Redraw the circuit in Figure 1 by replacing the delta abc with star network. Also mark the bottom delta as def .

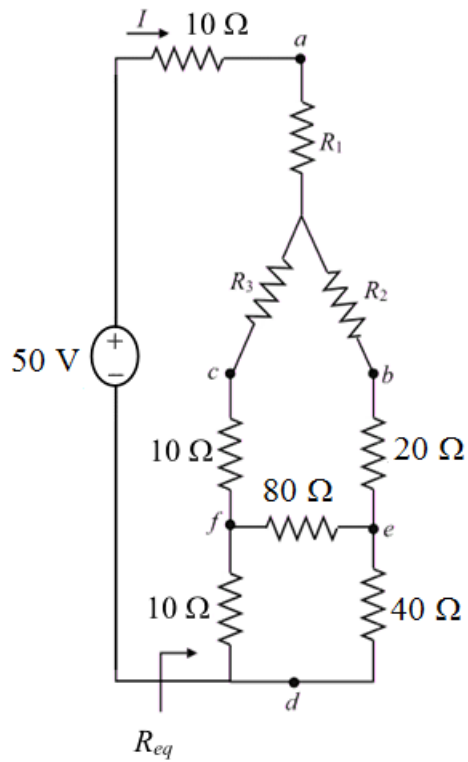


Figure 2

Step 3 of 7

Determine the equivalent Y resistances marked in the circuit shown in Figure 3.

$$\begin{aligned} R_1 &= \frac{(10)(30)}{10+10+30} \\ &= \frac{300}{50} \\ &= 6 \, \Omega \end{aligned}$$

$$\begin{aligned} R_2 &= \frac{(10)(30)}{10+10+30} \\ &= \frac{300}{50} \\ &= 6 \, \Omega \end{aligned}$$

$$\begin{aligned} R_3 &= \frac{(10)(10)}{10+10+30} \\ &= \frac{100}{50} \\ &= 2 \, \Omega \end{aligned}$$

Step 4 of 7

Redraw the circuit in Figure 2 by replacing the delta def with star network.

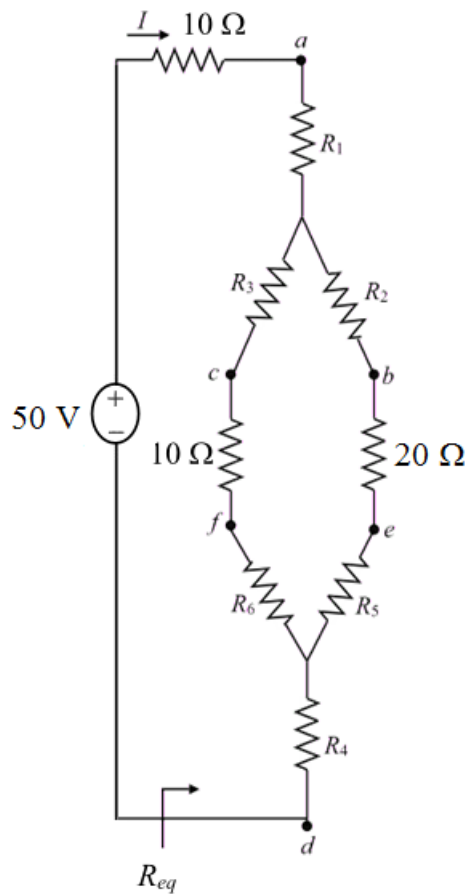


Figure 3

Step 5 of 7

Determine the equivalent Y resistances marked in the circuit shown in Figure 3.

$$\begin{aligned} R_4 &= \frac{(10)(40)}{10 + 40 + 80} \\ &= \frac{400}{130} \\ &= 3.08 \, \Omega \end{aligned}$$

$$\begin{aligned} R_5 &= \frac{(40)(80)}{10 + 40 + 80} \\ &= \frac{3200}{130} \\ &= 24.62 \, \Omega \end{aligned}$$

$$\begin{aligned} R_6 &= \frac{(10)(80)}{10 + 40 + 80} \\ &= \frac{800}{130} \\ &= 6.15 \, \Omega \end{aligned}$$

Step 6 of 7

Determine the equivalent resistance of the network in Figure 3.

$$\begin{aligned}R_{\text{eq}} &= 10 + R_1 + \left[(R_3 + 10 + R_6) \parallel (R_2 + 20 + R_5) \right] + R_4 \\&= 10 + 6 + \left[(2 + 10 + 6.15) \parallel (6 + 20 + 24.62) \right] + 3.08 \\&= 10 + 6 + \left[(18.15) \parallel (50.62) \right] + 3.08 \\&= 10 + 6 + \left[\frac{(18.15)(50.62)}{(18.15) + (50.62)} \right] + 3.08 \\&= 10 + 6 + 13.36 + 3.08 \\&= 32.44 \, \Omega\end{aligned}$$

Therefore, the equivalent resistance of the network is $\boxed{32.44 \, \Omega}$.

Step 7 of 7

Determine the current from the source, I .

$$\begin{aligned}I &= \frac{50}{R_{\text{eq}}} \\&= \frac{50}{32.44} \\&= 1.5413 \, \text{A}\end{aligned}$$

Therefore, the current delivered by the source, I is $\boxed{1.5413 \, \text{A}}$.